

LESSON 6-3 • SESSION 2 • PRACTICE • FORM B**Finding the Value of a Power**

Learning target – I can find the value of a power by multiplying the base the number of times the exponent shows. · 6.EE.A.1

REMEMBER

- Write the **expanded form**, then multiply one factor at a time.
- $4^3 = 4 \times 4 \times 4 = 64$, **not** $4 \times 3 = 12$.
- Decimals and fractions work the same way: $0.3^2 = 0.3 \times 0.3 = 0.09$.

WORKED EXAMPLE

Find the value of 3^4 .

Step 1 Expand: $3^4 = 3 \times 3 \times 3 \times 3$.

Step 2 Multiply: $3 \times 3 = 9$, $9 \times 3 = 27$, $27 \times 3 = 81$.

Step 3 Check: I used 3 as a factor 4 times.

Answer $3^4 = 81$

WORDS evaluate (*evaluar*) · squared (*al cuadrado*) · cubed (*al cubo*) · expanded form (*forma desarrollada*)

PRACTICE

show your work · label units

1. Find the value of each power.

a. $9^2 = \underline{\hspace{2cm}}$

b. $2^3 = \underline{\hspace{2cm}}$

c. $5^2 = \underline{\hspace{2cm}}$

d. $13^2 = \underline{\hspace{2cm}}$

e. $1.3^2 = \underline{\hspace{2cm}}$

f. $(2/5)^2 = \underline{\hspace{2cm}}$

2. **Power Table.** A garden club makes a table of powers of 3. Complete it. Then tell what each value is multiplied by to get the next one.

Power	Expanded form	Value
3^1		
3^2		
3^3		
3^4		

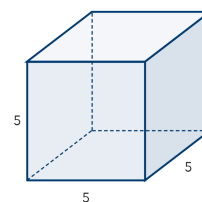
Multiply by _____

3. **Storage Cube.** A storage cube has edges of 5 units, so its volume is 5^3 cubic units. Find the volume.

Hint Use 5 as a factor three times.

Show your work.

Answer _____ cubic units



4. **Plant Cuttings.** A count of plant cuttings is multiplied by 6 each week for 3 weeks, so it is 6^3 . Evaluate it.

Hint $6 \times 6 = 36$, then multiply by 6 again.

Show your work.

Value _____

5. **Poster Claim.** A classroom poster says $8^2 = 16$. Justify or correct the poster.

Hint Expand the power into factors first.

Show your work.

Value _____

6. **Which Is Biggest?** Evaluate 3^4 , 4^3 and 2^7 . Order them from least to greatest.

Hint Evaluate each power before comparing.

Show your work.

Order _____

7. **Decimal Powers.** Evaluate each power.

Hint Count decimal places: 0.4×0.4 has two.

a. $0.4^2 =$ _____

b. $2.5^2 =$ _____

8. **Tile Stages.** A tile design has 4 times as many tiles at each new stage. It starts with 1 tile.

a. Write the number of tiles after 3 stages as a power of 4, then evaluate. _____ tiles

b. How many tiles after 4 stages? _____ tiles

c. How many more tiles after 4 stages than after 3 stages? _____ tiles

FIND THE ERROR

explain the mistake, then fix it

9. **Malik's Work.** Malik evaluated 9^3 . He wrote $9 \times 3 = 27$ and said the value is 27. Explain his mistake. Then find the correct value.

TEST PRACTICE

choose the best answer

10. 10^3 equals –

A 30

B 13

C 100

D 1,000

11. 6^2 equals –

A 12

B 8

C 36

D 66

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

15 Q	16 X	18 Z	6.25 O	64 C	216 A	125 F	256 R	192 S	0.16 T
#3	#4	#5	#7a	#7b	#8b	#8c			

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____